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Department of Electrical Engineering

EE-504: Power Electronics – Vivid Study Note SCR Protection & Commutation (Module 2)

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💡 **Welcome, Engineers!** Think of an SCR like a **heavy-duty door**.

- **Gate** = The key that *opens* the door (but **cannot close** it).
- **Anode-Cathode** = The door itself, carrying heavy current.
- **Protection Circuits** = Bodyguards that prevent the door from being kicked open accidentally (dv/dt) or damaged by a sudden rush of people (di/dt).
- **Commutation** = The process of *closing* the door (turning it OFF) when the key won't work.

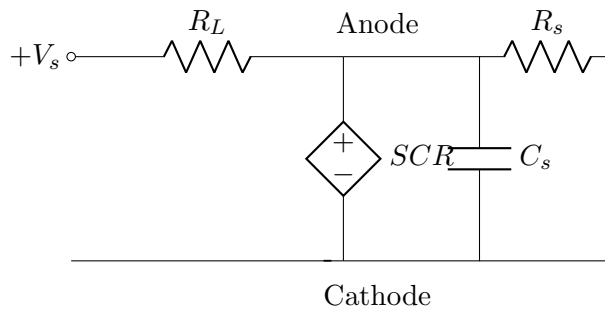
Let's dive into how these "bodyguards" and "closing mechanisms" work!

1 🛡️ Part 1: SCR Protection – The Bodyguards

1.1 The dv/dt Threat: The "False Trigger"

The Problem: When an SCR is OFF (blocking), it looks like a capacitor across junction J2. If the anode-cathode voltage (V_{AK}) rises too fast, a tiny current $i = C_{J2} \frac{dv}{dt}$ flows *into* the gate region. If this current is high enough, the SCR **turns ON by itself** — without any gate pulse! This is a **false trigger** and can cause short circuits.

The Solution: The Snubber Circuit ($R_s + C_s$) We place a capacitor (C_s) across the SCR. This capacitor **absorbs the sudden voltage change**, slowing down the dv/dt seen by the SCR.



Key Formula (Snubber Sizing): To limit dv/dt to a safe value, the snubber capacitor must be at least:

$$C_s \geq \frac{V_s}{(R_L + R_s) \cdot (dv/dt)_{max}}$$

- V_s = Supply voltage (Volts)

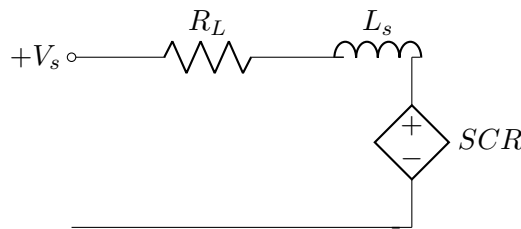
- R_L = Load resistance (Ω)
- R_s = Snubber resistance (Ω) (Usually chosen to damp oscillations)

Energy lost per switching: $E_s = \frac{1}{2} C_s V_s^2$ (This is the "cost" of protection!)

1.2 The di/dt Threat: The "Local Hotspot"

The Problem: When an SCR turns ON, conduction does not spread across the whole silicon wafer instantly. It starts near the gate. If the anode current (I_A) rises *too fast*, that tiny gate-adjacent area overheats and melts before the rest of the junction can turn on.

The Solution: The Series Inductor (L_s) An inductor opposes sudden changes in current. By placing a small inductor in series with the SCR, we give the current time to spread evenly across the junction.



Key Formula (Inductor Sizing):

$$L_s \geq \frac{V_s}{(di/dt)_{max}}$$

Where $(di/dt)_{max}$ is the maximum allowed current rise rate (given in the SCR datasheet, typically A/ μ s).

1.3 Gate Protection: Protecting the "Brain"

The gate-cathode junction is sensitive. Too much voltage or reverse voltage breaks it.

The 3 Musketeers of Gate Protection:

1. **Series Resistor (R_G):** Limits the peak gate current. (Protects against over-current).
2. **Gate-Cathode Resistor (R_{GK}):** Provides a path for leakage current to bypass the gate. (Prevents thermal runaway and false triggering).
3. **Zener Diode (Clamp):** Clamps the gate voltage to a safe level (e.g., 10V). (Protects against over-voltage spikes).

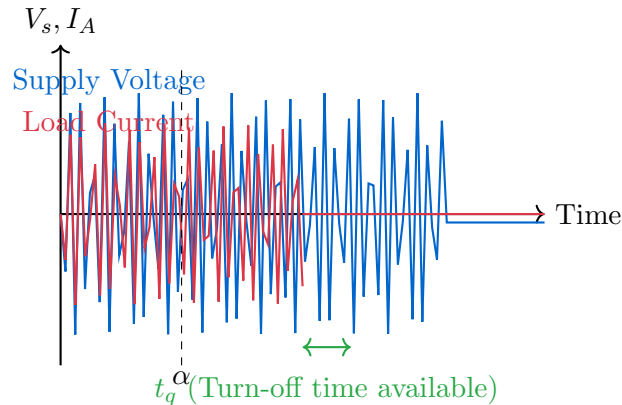
2 ⇌ Part 2: SCR Commutation – The "Goodbye" Process

The Golden Rule of SCRs: Once turned ON, the gate is **useless** for turning it OFF! The only way to turn OFF an SCR is to reduce the anode current (I_A) below the **Holding Current** (I_H). The process of forcing I_A to zero is called **Commutation**.

2.1 Natural Commutation: "Nature Does the Job"

Where it works: AC circuits (like rectifiers, AC voltage controllers).

How it works: The AC supply voltage naturally reverses polarity every half-cycle. When the voltage goes negative, the current tries to reverse. The SCR can only conduct in one direction, so I_A naturally drops to zero. It turns OFF on its own.



Full-Wave Rectifier Turn-Off Time:

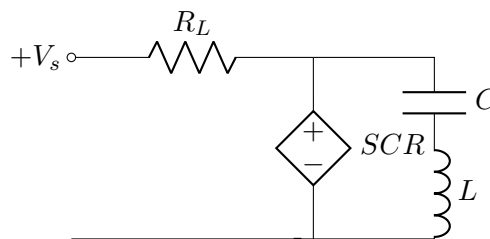
$$t_q = \frac{\alpha}{\omega}$$

Where α is the firing angle and $\omega = 2\pi f$. **Critical Condition:** For commutation to be successful, the circuit must provide $t_q > t_{off}$ (SCR recovery time). If $t_q < t_{off}$, the SCR will turn ON again immediately (**Commutation Failure**).

2.2 Forced Commutation: "We Do the Job"

Where it works: DC circuits (Choppers, Inverters). Since the DC supply never goes negative, we must create an **artificial reverse voltage** across the SCR.

How it works (Class A / Self-Commutated): We use a charged capacitor (C) and an inductor (L) to create a brief oscillating current. This current opposes the load current, forcing the net current through the SCR to zero, allowing it to turn OFF.



Forced Commutation Formulas (Class A):

- **Resonant Frequency:** $f_r = \frac{1}{2\pi\sqrt{LC}}$
- **Commutation Time:** $t_c = \pi\sqrt{LC}$ (Time SCR sees reverse voltage)
- **Peak Commutation Current:** $I_{peak} = V_s\sqrt{\frac{C}{L}}$

Critical Condition: The commutation time must exceed the SCR turn-off time:

$t_c > t_{off}$

3 The Ultimate Comparison: Natural vs Forced

Parameter	Natural Commutation	Forced Commutation
Supply Type	AC Supply (alternating)	DC Supply (constant)
External Circuit	None needed	L-C tank or auxiliary SCR
Converter Types	Rectifiers, AC Controllers	Choppers, Inverters
Complexity	Simple and Robust	Complex and Expensive
Power Factor	Depends on firing angle	Often requires leading VARs

Table 1: Natural vs Forced Commutation at a glance

4 Quick Revision Pillars (For the CL Activity)

✔ Before you walk into the CL class, make sure you remember these 5 "Pillars":

1. **Snubber** (R_sC_s) slows down dv/dt – It prevents false triggering.
2. **Series Inductor** (L_s) slows down di/dt – It prevents localized heating.
3. **Gate protection** uses R_G , R_{GK} , and **Zener** – To protect the fragile gate junction.
4. **Natural Commutation** uses **AC reversal** – Only works if the circuit turn-off time $t_q >$ SCR recovery time t_{off} .
5. **Forced Commutation** uses **L-C resonance** – To artificially force current to zero in DC circuits (Class A is the basic type).

⚠ Watch Out! (Common CL Mistakes)

- **Confusing C_s formula:** Don't flip it! $C_s = \frac{V_s}{(R_{total}) \cdot (dv/dt)}$.
- **t_q in rectifiers:** For a full-wave rectifier, $t_q = \alpha/\omega$. **Not** $(\pi - \alpha)/\omega$ (that's for half-wave).
- **Natural in DC:** You **cannot** use natural commutation in a DC chopper. The voltage never reverses!
- **L_s units:** di/dt is usually given in A/ μ s. Convert it to A/s before calculating Henrys (e.g., 50 A/ μ s = 50×10^6 A/s).

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